Language Models for Rational, Evidence-Driven Trading

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1 Problem statement

Forecasting financial markets is a longstanding challenge due to the high volatility and non-stationary nature of asset prices, which are influenced by both structured numerical factors and unstructured textual information. Traditional statistical or machine learning models often process historical price signals and market indicators, but struggle to incorporate complex, evolving textual inputs such as news headlines, policy speeches, and social media sentiment. Conversely, large language models (LLMs) demonstrate strong semantic reasoning capabilities over text, yet are not inherently designed to work with high-frequency numerical data or to produce transparent predictions in financial settings.

A major limitation of current approaches is the absence of models that are simultaneously (i) accurate, (ii) interpretable, and (iii) capable of integrating multimodal information. Furthermore, the lack of faithful explanations—particularly in algorithmic trading systems—raises concerns about trust, compliance, and human-in-the-loop decision making.

In this work, we propose a hybrid architecture that combines:

- **Structured numerical features**, extracted from global asset prices (e.g., XAU-USD, oil, and stock indices) at 4-hour intervals with technical indicators such as SMA, EMA, and MACD;
- **Semantic textual features**, extracted from Reddit, Twitter, and economic news via LLM-based clustering and transformed into interpretable binary feature vectors;
- **Two complementary prediction streams**: one powered by XGBoost over fused structured and textual features, and one directly using LLM prompts over human-readable textual summaries;
- **A two-tier explanation layer**, where SHAP values derived from the XGBoost model and raw text are passed to an LLM to generate natural language justifications, alongside self-explanations of LLM-based predictions.

Our goal is to develop a system that not only achieves strong predictive performance, but also produces rationales that are human-interpretable, diverse, and faithful to the model's internal reasoning. To assess this, we will employ a range of metrics including ROUGE, BLEU, BERTScore, FMR (Feature Matching Rate), and human-aligned sentiment coherence measures.

Ultimately, we aim to demonstrate that combining structured modeling, LLM-based textual reasoning, and explainability tools enables the creation of a rational, evidence-driven trading agent that balances profitability with interpretability.

2 What you proposed vs. what you accomplished

- ~~Collect structured numerical data (gold, oil, major stocks) at 4-hour intervals with aligned timestamps~~
- ~~Collect unstructured textual data from Reddit, Twitter, and financial news~~
- <u>Preprocess and clean textual data and extract prompt-ready blocks for LLM labeling</u>:
  In progress. Text scraping is complete, but LLM-compatible formatting and noise reduction is ongoing due to heterogeneity of sources.
- <u>Use LLMs to extract high-impact features from text and generate one-hot encoded</u>

representations:
Not yet started. Will begin immediately after preprocessing phase.

- Cluster textual features to reduce dimensionality and semantic redundancy:
Not started yet.

- Compute technical indicators (SMA, EMA, MACD, etc.) on numerical data:
Partially complete. Feature engineering functions have been written; application to the full dataset is pending.

- Train XGBoost on combined numerical and textual features to predict buy/hold/sell signals:
Not yet started.

- Feed prompt-based summaries into LLMs to generate predictions and explanations:
Not yet started.

- Generate explanations for XGBoost via SHAP and feed them to LLM for natural language rationales:
Not yet started.

- Evaluate explanations using automatic metrics (ROUGE, BLEU, FMR, etc.) and human-in-the-loop feedback:
Not yet started.

3 Related work

3.1 LLMs for Financial Forecasting and Sentiment Understanding

The application of large language models (LLMs) to financial forecasting has received growing attention in recent years, driven by their capacity to capture semantics in textual signals. However, the inherently volatile and non-stationary nature of financial markets poses unique challenges (Liu and Wang, 2023; Zhang and Li, 2024; Li et al., 2023a). Recent work demonstrates that fine-tuning pretrained LLMs with domain-specific corpora such as financial news or analyst reports can improve forecasting accuracy (Guo and Hauptmann, 2024; Fu et al., 2024; Xiao et al., 2025a; Liu et al., 2023). Notably, Guo and Hauptmann (2024) enhanced stock return predictions by using token-level embeddings from financial news, showing that textual representation learning can complement traditional quantitative models.

Efforts to bridge the gap between textual and numerical data have led to hybrid models that jointly encode market indicators and natural language context (Wang and Sun, 2023; Chen and Huang, 2024). For example, Fu et al. (2024) introduced a framework that tightly couples newsflow-based embeddings with time-series forecasting heads. Similarly, Huang and Zhou (2024) proposed StockLLM, a retrieval-augmented generation model fine-tuned for financial domains, achieving superior performance in mid-term forecasting tasks.

Parallel to this trend, financial sentiment analysis has become a cornerstone for LLM-based forecasting. Studies such as Bollen et al. (2011) and Chakraborty and Basu (2024) have shown strong links between social media mood and market behavior. Models that incorporate both textual sentiment and price trends—such as those using BERT-LSTM hybrids—offer a more nuanced view of market sentiment over time (Xu and Li, 2023; Chen and Huang, 2024). These works highlight the importance of temporal alignment and multimodal integration in high-frequency forecasting.

3.2 LLM Agents, Market Simulation, and Explainability

Beyond static predictions, LLMs have recently been deployed as autonomous trading agents in simulated environments. For instance, StockAgent (Zhang et al., 2023) introduces a multi-agent architecture where LLMs emulate investor profiles within a synthetic market. Building on this idea, frameworks such as TradingAgents (Xiao et al., 2025b) enable agents to dynamically assume roles like market makers or momentum traders (Li et al., 2021), facilitating competitive or cooperative trading strategies in interactive settings.

More sophisticated agent-based systems, like FinMem (Yu et al., 2023) and TradingGPT (Li et al., 2023b), incorporate persistent memory and instruction tuning to support adaptive reasoning across trading episodes. These systems push the boundaries of LLMs as rational economic actors, capable of scenario-aware planning and portfolio optimization.

A crucial challenge in deploying such systems lies in interpretability. As financial decisions become more automated, ensuring transparency and explainability becomes imperative for compliance and trust (Arrieta and et al., 2020; Yeo et al., 2023). Classic model-agnostic tools like SHAP (Lundberg and Lee, 2017) and LIME (Ribeiro et al., 2016) have been widely adopted to provide post-hoc explanations of model outputs. Recent work also explores integrating interpretability into the generative process of LLMs themselves (Miller, 2019; Carvalho et al., 2019), offering more faithful and human-aligned justifications.

Our work builds on this foundation by combining LLM-based feature extraction, hybrid modeling, and a two-layer explanation mechanism using SHAP and natural language rationales. Unlike prior work, we explicitly align text-derived signals with fixed prediction intervals and evaluate explanations using both automatic metrics and financial performance outcomes.

4 Your Dataset

Our dataset consists of two complementary modalities: structured numerical data representing financial market activity, and unstructured textual data capturing investor sentiment and global economic events. These two data streams are aligned on a fixed 4-hour interval, providing a high-frequency multimodal snapshot of the market state.

Textual data. We collect unstructured text from sources such as TradingView, Twitter, Reddit, financial news websites, and official policy announcements (Bollen et al., 2011; Morstatter et al., 2013). This content includes political speeches (e.g., “Trump delivered a trade policy speech”), analyst commentaries, and social media discussions of cryptocurrencies and stocks. These texts are concatenated into a single prompt and passed to an LLM, which is asked to extract and summarize major impactful events using short phrases (e.g., “rate hike”, “supply shock”, “SEC lawsuit”).

These LLM-extracted phrases are clustered using semantic similarity (e.g., via cosine distance in embedding space), resulting in a reduced feature space of interpretable event clusters. For each 4-hour time slice, we construct a binary vector indicating which clusters are active.

Numerical data. We obtain high-frequency price data (OHLCV: Open, High, Low, Close, Volume) from multiple global markets including gold (XAU-USD), oil, and major equities. From these, we compute engineered indicators such as SMA, EMA, and MACD, which are common in algorithmic trading (Tsantekidis et al., 2017). These features are aligned with the same 4-hour intervals as the text-derived clusters.

Basic statistics. At the current stage, we have collected:

- Over 2,000 4-hour intervals of synchronized data from 2023–2024.
- Over 15,000 unique textual inputs across Reddit, Twitter, and news.
- $\sim$60 LLM-generated semantic features clustered into $\sim$10 high-level event types.
- 35 numerical indicators per time slice, computed across different windows and assets.

Example input/output pairs. A simplified example of one training instance is:

- **Input:**
  - Text features: [“rate hike”, “inflation fear”, “G7 summit”]
  - Numerical features: [EMA_{12} = 0.94, SMA_{50} = 1.02, ...]
- **Output:** 2 (Sell)

4.1 Data Preprocessing

Textual data undergoes several preprocessing steps: language filtering (English-only), noise removal (e.g., hashtags, mentions, URLs), and prompt formatting. We concatenate all content within a 4-hour window and send it to an LLM in a templated prompt. The LLM’s short output phrases are extracted and lemmatized, then mapped to pre-defined clusters. For numerical data, we normalize all technical indicators and handle missing values using forward-fill interpolation.

4.2 Data Annotation

Our project does not rely on human annotation in the conventional sense. Instead, we use LLMs to automatically generate semantic labels from text and derive binary features from these outputs. However, we manually reviewed $\sim$100 examples to verify that the LLM-generated features are coherent and semantically aligned. Agreement across team members was high, though we noticed some ambiguity in cases where LLM outputs were too generic (e.g., “market fear”) or domain-specific (e.g., "Grayscale ruling")—these were subsequently handled via manual cluster assignment.

5 Baselines

Our experimental setup includes two primary models that operate over the same core feature set but use fundamentally different representations and modalities. This contrast allows us to evaluate the impact of (i) using LLMs versus traditional models, and (ii) using raw natural language prompts versus binary vectorized inputs.

1. XGBoost with numerical and one-hot textual features. The first model is a gradient boosting classifier (XGBoost) trained on a fused feature vector composed of:

- Numerical indicators extracted from OHLCV data (e.g., SMA, EMA, MACD),
- One-hot encoded textual features derived from LLM-generated event summaries clustered into a fixed vocabulary.

This model represents a classical supervised approach where all input features are structured and fed into a tabular learning framework. The one-hot encoding provides sparse binary indicators of event occurrence in each 4-hour window.

2. LLM predictor with prompt-based semantic features. In contrast, our second model leverages a large language model (LLM) that receives the entire feature set in a human-readable, prompt-formatted textual description. For example, the input prompt may include: `''Recent events: rate hike, inflation fear, OPEC cuts. Technical indicators: gold SMA rising, MACD diverging.''` The LLM is then asked to predict one of three labels: buy, sell, or neutral. This setting exploits the model's capacity for semantic reasoning and pattern understanding in natural language.

5.1 Train/Test Split and Settings

We use a chronological split of 70/15/15% for train/validation/test over the sequence of 4-hour intervals. The entire feature extraction and clustering process is performed solely on the training data to prevent temporal leakage. The same samples are used for both models to ensure fairness.

5.2 Why These Baselines?

We chose these two configurations to isolate the impact of representation format on model performance. Both models receive the same market signals, but only the LLM has access to natural language descriptions and semantic abstraction. This enables us to test whether a reasoning-based model with language understanding can outperform a structured model that relies on engineered inputs and statistical correlations.

6 Your approach

7 Your Approach

Our approach builds a dual-model hybrid architecture combining the structured modeling power of XGBoost with the reasoning capabilities of large language models (LLMs). After preparing both numerical and textual features, we feed all inputs into two parallel prediction engines:

1. A gradient boosting classifier (XGBoost) trained on engineered numerical indicators and one-hot encoded textual event clusters;
2. A prompt-based LLM that receives the same features in natural language form and generates a discrete prediction (buy, sell, or neutral) along with a free-text explanation.

7.1 Explainability Pipeline

Once the XGBoost model outputs its decision, we use the SHAP (SHapley Additive exPlanations) framework to extract local feature attributions per instance. These SHAP values, along with the raw event text from which the features were derived, are passed to the LLM. The LLM then generates an explanation that is both informed by the model's internal logic (via SHAP) and grounded in human-readable content. This explanation pipeline ensures that the rationale remains interpretable, faithful, and actionable for end-users.

Moreover, the LLM model used for prediction is also required to self-explain its decision. In other words, we compare not only performance across models, but also the clarity and utility of their justifications.

7.2 Roles of the LLM

The LLM is employed in three distinct capacities:

- **Feature extraction:** summarizing and labeling events from raw text,
- **Decision prediction:** inferring the correct trading action based on textual and numerical context,
- **Explanation generation:** producing natural language justifications for both its own outputs and those of the XGBoost model.

7.3 Evaluation of Explanations

We implement a comprehensive evaluation framework for generated explanations using both automatic metrics and qualitative judgments. Standard n-gram overlap metrics such as ROUGE (Lin, 2004) and BLEU (Papineni et al., 2002) are used to assess lexical fidelity, while METEOR (Banerjee and Lavie, 2005) adds a semantic layer via synonymy and reordering.

For deeper semantic alignment, we use BERTScore (Zhang* et al., 2020), and to balance generation quality versus diversity, we apply MAUVE (Pillutla et al., 2021). Other dimensions include:

- **Factual coherence:** Feature Matching Rate (FMR) and Feature Coverage Rate (FCR) (Cheng et al., 2023),
- **Diversity:** Unique Sentence Ratio (USR), DISTINCT, and Entropy (ENTR) (Li et al., 2020; Wang et al., 2022),
- **Sentiment consistency:** Human Sentiment Coherence (HSC) and Automated Sentiment Coherence (ASC) (Dong et al., 2017; Hada and Shevade, 2021),
- **Logical soundness:** entailment-based metrics and Attribute Classification Performance (ACP) (Xie et al., 2023).

7.4 Profitability-Aware Assessment

In addition to standard classification metrics (F1, accuracy), we define a custom evaluation based on simulated profitability—i.e., the cumulative gain or loss accrued by following model-generated predictions over time. We further investigate how the quality of explanations (e.g., FMR or BERTScore) correlates with the financial utility of the decision.

7.5 Implementation and Environment

We implemented all code in Python using:

- `scikit-learn` and `xgboost` for gradient boosting,
- `shap` for interpretability analysis,
- `transformers` and `langchain` for LLM interaction,
- Prompt-based access to LLMs such as DeepSeek or GPT-3.5 (via API).

We currently use Google Colab Pro with NVIDIA T4 GPUs for LLM-based inference. We also employ memory-mapping and batch-level inference to handle large volumes of prompts efficiently. No fine-tuning is applied; all predictions are zero-shot or few-shot via prompt engineering.

7.6 Current Status and Challenges

We have completed dataset construction and are finalizing the prompt formats and SHAP-to-text explanation pipeline. The XGBoost model is trained and integrated. The LLM pipeline is partially deployed, though latency and token-length limits in Colab environments occasionally force prompt truncation or batching adjustments.

No major bugs have blocked our progress so far, although balancing token limits with explanation granularity remains an active design challenge.

8 Error analysis

What kinds of inputs do your baselines fail at? What about your approach? Are there any semantic or syntactic commonalities between these difficult examples? **We would like to see a manual error analysis (e.g., annotate 100-200 failed examples for various properties, and then discuss the results and hypothesize about why the).**

9 Contributions of group members

List what each member of the group contributed to this project here. For example:

- member 1: did data collection / processing and lots of writing
- member 2: built and trained models
- member 3: error analysis and annotations

If you would like to privately share more information about the workload division that may have caused extenuating circumstances (e.g., a member of the group was unreachable and did no work), please send a detailed note to the instructors GMail account. We will take these notes into account when assigning individual grades.

10 Conclusion

You've now gotten your hands dirty with NLP tools and techniques! What takeaways do you have about your project? What proved surprisingly difficult to accomplish? Were your surprised by your results? If you could continue working on your project in the future, what directions would you pursue?

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